

ASSIGNMENT 01 – PROJECT PROPOSAL

MindSpring – Smart Mental Wellness and Habit Tracking Mobile Application

Field	Details
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Registration Number	IT22235262
Last Digit / Allocated Topic	2 – Smart Mental Wellness and Habit Tracking Mobile Application
Proposed Application Name	MindSpring
Platform / Language	Android (Kotlin, Android Studio)
Submission	Phase 01 – Planning & Requirement Analysis

1. Introduction

MindSpring is a proposed Android mobile application that combines daily mood journaling with personal habit tracking in a single, privacy-focused app. The core idea is that mental wellbeing is strongly influenced by everyday routines such as sleep, physical activity, hydration and screen time, yet most people never see the relationship between the two because these things are tracked separately, if at all.

By recording both a short daily mood check-in and the completion of user-defined habits, MindSpring can present simple visual insights that show the user which routines tend to accompany their better days. The application will be developed in Kotlin using Android Studio and will follow the Software Development Life Cycle stages of Planning, Requirement Definition, Design, Development and Testing.

2. Problem Statement

Stress, anxiety and low mood are increasingly common among university students and young working adults, but the tools available to manage them are often unsuitable for everyday use:

- Mood and routine are tracked in isolation. Journaling apps capture feelings but ignore behaviour, while habit apps count streaks but ignore how the user actually feels.
- Users receive data, not insight. Many apps display raw logs and leave the user to interpret them, so the connection between habits and wellbeing is never made visible.

- Barriers to consistent use. Popular wellness apps frequently lock meaningful features behind subscriptions, require constant internet connectivity, or demand lengthy daily entries that users abandon within a week.
- Privacy concerns. Mood data is deeply personal, and users are reluctant to store it on third-party cloud servers without a clear reason.

There is therefore a need for a lightweight, offline-capable application that makes a daily check-in take less than a minute and then translates that data into a clear, understandable picture of the user's wellbeing over time.

3. Proposed Solution

MindSpring addresses the problem through four connected components:

- A quick daily mood check-in using a five-point scale, selectable emotion tags and an optional short note.
- A habit tracker where users create their own habits, set a target frequency, mark them complete and build streaks.
- An insights dashboard that charts mood over time and highlights which habits are most often completed on higher-mood days.
- Two guided micro-tools — a timed breathing exercise and a gratitude prompt — that give the user an immediate action to take, not just a record to read.

All data is stored locally on the device using a Room database, which keeps personal information private and allows the application to work fully offline.

4. Target Audience

Audience	Description	Need Addressed
Primary: University students (18–25)	Students managing academic pressure, irregular sleep and inconsistent routines.	A fast, free way to notice stress patterns and build stable daily habits.
Secondary: Young working adults (25–35)	Early-career professionals coping with workload and work–life balance.	Lightweight self-monitoring that fits into a busy schedule.
Tertiary: General self-improvement users	Anyone building habits such as exercise, reading, hydration or reduced screen time.	Habit streaks with visible feedback on how routines affect mood.

The application is designed as a self-awareness and wellbeing tool for the general public. It is not intended for clinical use and does not target users seeking diagnosis or treatment.

5. Feasibility Study

5.1 Technical Feasibility

The application requires no specialised hardware or external paid services. Kotlin, Android Studio, Jetpack Compose and the Room persistence library are all freely available and well documented. Charting can be implemented using Compose Canvas or an open-source charting library, and reminders can be scheduled with WorkManager and the Android notification framework. All required technologies fall within the scope of the SE4041 module.

5.2 Economic Feasibility

Development cost is effectively zero. Android Studio and the Android SDK are free, testing can be carried out on the Android Emulator or a personal device, and GitHub provides free repository hosting. Because data is stored on-device, there are no server or database hosting costs.

5.3 Operational Feasibility

The intended users are already comfortable with smartphone applications, and the interaction model — tapping a mood, ticking a habit — is deliberately simple. The daily check-in is designed to take under one minute, which makes sustained use realistic.

5.4 Legal and Ethical Feasibility

Because the application handles sensitive personal information, all data will remain on the user's device and will not be transmitted to any external server. MindSpring will not provide diagnosis, clinical screening scores or medical advice. A clear in-app disclaimer will state that the application is a self-reflection tool and will direct users to professional support services if they record sustained low mood.

5.5 Schedule Feasibility

Phase	Main Activities	Duration
Planning & Requirement Analysis	Problem study, scope definition, requirement gathering	Week 1–2
Design	Figma UI prototype, colour scheme, navigation flow	Week 3–4
Development	Kotlin implementation, Room database integration	Week 5–9
Testing	Unit testing, integration testing, defect fixing	Week 10–11
Documentation	Final report, screenshots, GitHub finalisation	Week 12

6. Scope of the Application

6.1 In Scope

- User registration and login with a locally stored profile.

- Daily mood check-in with rating, emotion tags and optional note.
- Creation, editing and deletion of custom habits with target frequency.
- Daily habit completion tracking with streak calculation.
- Insights dashboard with a mood trend chart and habit completion summary.
- Guided breathing exercise and gratitude journal.
- Local reminder notifications for check-ins and habits.
- Local data persistence using the Room database.
- Settings including dark mode and data reset.

6.2 Out of Scope

- Clinical diagnosis, screening instruments or any form of medical advice.
- Live chat or consultation with therapists or counsellors.
- Cloud synchronisation across multiple devices.
- Integration with wearable devices or third-party health platforms.
- Social networking, community feeds or content sharing.
- iOS or web versions of the application.

7. Functional Requirements

ID	Requirement
FR-01	The system shall allow a user to register an account and log in to access their data.
FR-02	The system shall allow a user to record a daily mood entry with a rating, emotion tags and an optional note.
FR-03	The system shall allow a user to view, edit and delete previous mood entries.
FR-04	The system shall allow a user to create custom habits with a name, category and target frequency.
FR-05	The system shall allow a user to mark habits as complete for the current day and shall calculate the resulting streak.
FR-06	The system shall display a dashboard summarising today's mood, habit progress and current streaks.
FR-07	The system shall generate an insights view showing mood trends over a selected period and habit completion rates.
FR-08	The system shall provide a guided breathing exercise with a visual timer.
FR-09	The system shall provide a gratitude journal allowing daily positive entries.
FR-10	The system shall send local notification reminders for the daily check-in and for scheduled habits.

ID	Requirement
FR-11	The system shall store all user data persistently in a local Room database.
FR-12	The system shall allow the user to switch between light and dark themes and to clear all stored data.

8. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Usability	A daily mood check-in shall be completable in under one minute and within a maximum of three taps.
NFR-02	Performance	Screens shall load within two seconds and database operations shall run off the main thread using Kotlin Coroutines.
NFR-03	Reliability	The application shall not lose recorded data when closed unexpectedly, and all entries shall be committed immediately on save.
NFR-04	Security & Privacy	All personal data shall remain on the device and shall not be transmitted to any external server.
NFR-05	Availability	All core features shall function without an internet connection.
NFR-06	Maintainability	The application shall follow the MVVM architecture with a clear separation between UI, ViewModel and data layers.
NFR-07	Compatibility	The application shall support Android 8.0 (API 26) and above.
NFR-08	Accessibility	Colour contrast shall meet readability guidelines and the interface shall support system font scaling.

9. Key Features and Functionalities

Feature	Description
Daily Mood Check-in	A five-point mood scale with emotion tags and an optional note, recorded once per day.
Habit Builder	User-defined habits with categories, target frequency and daily completion tracking.
Streaks & Progress	Automatic streak counting and weekly completion percentages to encourage consistency.
Insights Dashboard	Mood trend chart over time alongside habit completion data, showing which routines accompany better days.
Breathing Exercise	A guided timed breathing session with an animated visual cue.

Feature	Description
Gratitude Journal	A short daily prompt for recording positive reflections.
Smart Reminders	Local notifications scheduled through WorkManager for check-ins and habits.
Personalisation	Light and dark themes, editable profile and full data reset.

10. Proposed Technology Stack

Component	Technology	Justification
Language	Kotlin	Official Android language, required by the assignment.
IDE	Android Studio	Standard Android development environment.
UI Toolkit	Jetpack Compose	Declarative UI with efficient state handling for reactive charts and lists.
Architecture	MVVM	Separates UI from business logic and improves testability.
Database	Room (SQLite)	Local, offline-first persistence satisfying the privacy requirement.
Concurrency	Coroutines & Flow	Keeps database work off the main thread and updates the UI reactively.
Notifications	WorkManager	Reliable scheduling of reminders across device restarts.
Testing	JUnit, MockK, Espresso	Supports both unit testing and integration testing.
Version Control	Git & GitHub	Public repository with regular commits as required.

11. UI Design Approach

The interface will follow the 60-30-10 colour rule to create a calm and consistent visual identity. Approximately 60% of each screen will use a soft off-white or pale sage background, 30% will use a deep teal for cards, surfaces and navigation elements, and the remaining 10% will use a warm amber accent reserved for primary actions, streak highlights and selected states. A complete interactive prototype will be produced in Figma before development begins.

12. Expected Outcome

On completion, MindSpring will be a fully functional Android application that allows a user to track their mood and habits daily, view meaningful insights connecting the two, and access simple wellbeing exercises — all while keeping their data private on the device. The project will be supported by a Figma prototype, a public GitHub repository containing the full development history, documented unit and integration test results, and a final report covering all completed SDLC stages.